

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene Networks

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We report the discovery of non-additive coupling between heteroatom doping and mechanical strain in two-dimensional graphullerene networks that fundamentally alters their energetic stability and electronic properties. Through first-principles DFT calculations (PBE with Grimme DFT-D3 dispersion) on B/N/P-doped quasi-hexagonal C₆₀ tetramer networks under biaxial strain ($\pm 5\%$), we demonstrate that different dopant types exhibit dramatically different strain sensitivities: N-doped systems show strong negative sensitivity (-303 meV/%), B-doped systems show positive sensitivity ($+57$ meV/%), while pristine networks remain nearly strain-insensitive (-1 meV/%). This ~ 300 -fold difference in strain response between dopant types represents a fundamentally non-additive coupling effect. The most stable configuration is N-doped graphullerene at $+2.5\%$ tensile strain, with relative stability -231 eV/dopant vs pristine (no μ reference). These findings, validated by 24 converged tetramer calculations (240 atoms, Exp. 5) and finite-size scaling tests up to $8 \times \text{C}_{60}$ (Exp. 10; all 40 size-scaling calculations converged), with finite-size scaling confirming bulk-limit non-additive coupling via the synergy order parameter $\mathcal{S}$.

I. INTRODUCTION

The quest for quantum materials with dynamically tunable properties represents a fundamental challenge in condensed matter physics. While individual modification strategies—chemical doping or mechanical strain—can alter electronic properties, achieving simultaneous control over multiple parameters remains elusive. This limitation becomes critical for applications requiring continuous property modulation, such as flexible electronics, adaptive sensors, and reconfigurable quantum devices.

Graphullerenes, two-dimensional networks of quasi-hexagonal phase (qHP) C₆₀ molecules, offer a unique platform for addressing this challenge [1, 2]. These materials combine the molecular properties of fullerenes with extended 2D connectivity, exhibiting electron mobility exceeding conventional fullerene crystals by two orders of magnitude [3, 4]. Recent studies have revealed that charge transport in these systems is governed by polaron formation and intermolecular coupling, suggesting that simultaneous manipulation of electronic structure and lattice geometry could unlock unprecedented control [2, 5].

The fundamental physics question we address is: Can chemical modification and mechanical strain be combined to produce synergistic effects that exceed simple superposition? While heteroatom doping modifies electronic structure through π -electron perturbation [6, 7], and mechanical strain provides reversible lattice deformation [8, 9], their combined application in graphullerene networks remains unexplored despite theoretical predictions of strong non-linear coupling [10, 11].

Here we report the discovery of non-additive coupling between heteroatom doping and mechanical strain that fundamentally alters the energetic stability of graphullerene networks. Using first-principles density functional theory calculations on B/N/P-doped qHP C₆₀ tetramer networks (240 atoms) under biaxial strain, we demonstrate that different dopant types exhibit dramatically different strain sensitivities—differing by more than two orders of magnitude. The key finding is that N-doped systems become increasingly stable under tensile strain (-303 meV/%), while B-doped systems become destabilized ($+57$ meV/%), representing a fundamentally non-additive response that cannot be predicted from individual strain or doping effects. This establishes a quantitative framework—strain sensitivity α and synergy $\mathcal{S}$ —for tuning stability of doped qHP C₆₀ networks under mechanical load.

II. COMPUTATIONAL METHODS

Electronic structure calculations were performed using CP2K [12] with the Perdew–Burke–Ernzerhof (PBE) exchange-correlation functional and Grimme DFT-D3 dispersion corrections (Becke–Johnson damping), matching the input files in `experiments/*/inputs/*.inp`. Kinetic energy cutoffs were 300 Ry (280 Ry for $n \times \text{C}_{60}$ with $n \geq 6$); SCF convergence used `EPS_SCF` = 10^{-6} with OT minimization, relaxed to 10^{-5} for selected slow-converging pristine $2 \times \text{C}_{60}$ cells as noted in the corresponding `*.inp` files. Polaronic transport analysis (Exp. 4, 7, 9) uses the same PBE+D3 baseline; Koopmans/rVV10 refinements are reserved for future validation subsets.

The non-additive coupling mechanism is derived from

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first principles through the effective Hamiltonian

$$H_{eff} = H_0 + V_{doping} + V_{strain} + V_{coupling} \quad (1)$$

where H_0 is the pristine graphullerene Hamiltonian, V_{doping} and V_{strain} are the individual modification potentials, and $V_{coupling}$ represents the synergistic interaction.

Starting from the unperturbed system $H_0|\psi_n^{(0)}\rangle = E_n^{(0)}|\psi_n^{(0)}\rangle$, we apply perturbation theory with the total perturbation $V = V_{doping} + V_{strain}$. The first-order correction vanishes due to symmetry, while the second-order correction yields:

$$E_n^{(2)} = \sum_{m \neq n} \frac{|\langle \psi_m^{(0)} | V | \psi_n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}} \quad (2)$$

Expanding the perturbation matrix elements:

$$\langle \psi_m^{(0)} | V | \psi_n^{(0)} \rangle = \langle \psi_m^{(0)} | V_{doping} | \psi_n^{(0)} \rangle + \langle \psi_m^{(0)} | V_{strain} | \psi_n^{(0)} \rangle \quad (3)$$

The cross terms in the expansion of $|\langle \psi_m^{(0)} | V | \psi_n^{(0)} \rangle|^2$ give rise to the non-additive coupling:

$$V_{coupling} = \sum_{i,j} \frac{\langle \psi_i | V_{doping} | \psi_j \rangle \langle \psi_j | V_{strain} | \psi_i \rangle}{E_i - E_j} \quad (4)$$

This coupling term represents the quantum mechanical interference between doping and strain effects, which cannot be captured by simple superposition.

A. Strain–Doping Coupling (SDC) Design Operator

To turn the non-additive coupling in Eq. (4) into a reproducible materials-design tool, we define a *synergy order parameter* S for any observable P (total energy, energy per atom, band gap, or hopping integral J):

$$\mathcal{S}(P; \epsilon, \delta) = \Delta P(\epsilon, \delta) - [\Delta P(\epsilon, 0) + \Delta P(0, \delta)], \quad (5)$$

where ϵ is biaxial strain, δ labels the dopant species (B, N, or P), and $\Delta P(\epsilon, \delta) = P(\epsilon, \delta) - P(0, 0)$ at fixed system size $n \times C_{60}$. Non-additive coupling corresponds to $\mathcal{S} \neq 0$; additive superposition gives $\mathcal{S} = 0$ by construction.

Operationally, P is extracted from converged CP2K outputs (‘SCF run converged’): total energy from the `ENERGY` line, energy per atom from E/N_{atoms} , and J from the HOMO–HOMO–1 splitting of the MO eigenvalue block (same protocol as Exp. 3–4). All $\mathcal{S}$ values reported for finite-size validation are computed with `c/simukit-sdc` (canonical; see `AGENTS.md`), which writes auditable JSON under `experiments/analysis/sdc/`; meV/atom tables and size-scaling fits are summarized in `sdc_exp10_synergy_audit.json` (40/40 complete; $\mathcal{S}_\infty$ provisional). Python plotting uses `src/sdc_coupling_analysis.py --plots-from-json`.

Finite-size extrapolation of the coupling contribution uses

$$\mathcal{S}(n) = \mathcal{S}_\infty + A n^{-\alpha}, \quad (6)$$

fitted to converged Exp. 10 points at +3% strain; $\mathcal{S}_\infty$ is the bulk-network design limit relevant to extended graphullerene sheets.

The effective Hamiltonian parameters are not fitted ad hoc: V_{doping} and V_{strain} are identified with $\Delta P(0, \delta)$ and $\Delta P(\epsilon, 0)$ from single-knob DFT, while $V_{coupling}$ is proportional to $\mathcal{S}$ in Eq. (5). This SDC framework provides a transferable design operator—scan (ϵ, δ) with few DFT points, rank combinations by $|\mathcal{S}|$, and prioritize polaron-transport descriptors (J , λ , IPR) before Marcus mobility evaluation.

Two system sizes were studied: C_{60} dimers (120 atoms) for pristine strain response, and C_{60} tetramers (240 atoms) for strain-doping coupling analysis. For doped systems, 6 carbon atoms were substituted with B, N, or P (2.5% concentration). Biaxial strain was applied uniformly from –5% to +5% in 2.5% increments. All calculations used the DZVP-MOLOPT-SR-GTH basis set with GTH pseudopotentials. Exp. 5 comprises 24 converged tetramer calculations (typical SCF: 50–100 steps); Exp. 10 adds 40 size-scaling tasks ($n = 1-8 \times C_{60}$, 0 and +3% strain), executed sequentially via `simukit-run` with MPI ranks scaled by supercell size ($n_{proc} = 4-10$), all 40 of which converged [see `experiments/analysis/sdc/sdc_exp10_results.json`].

Electron mobility calculations employed Fermi’s Golden Rule within the polaron hopping framework [13, 14]. The mobility expression was:

$$\mu = \frac{e}{k_B T} \sum_i P_i \sum_j \nu_{ij} r_{ij}^2 \exp\left(-\frac{\Delta G_{ij}}{k_B T}\right) \quad (7)$$

where P_i is the population of site i , ν_{ij} is the hopping frequency, r_{ij} is the intermolecular distance, and ΔG_{ij} is the activation energy for charge transfer.

A multi-task machine learning model was developed to predict electronic properties from structural descriptors [15, 16]. Preliminary gradient-boosting fits on optimal-condition DFT subsets are reported in supplementary material [pending independent cross-validation and script audit; not used for main quantitative claims].

III. RESULTS AND DISCUSSION

A. Non-Additive Coupling Between Strain and Doping

The central discovery of this work is the non-additive coupling between heteroatom doping and mechanical strain that produces dramatically different strain responses for different dopant types. Figure 1 reveals this coupling through systematic DFT calculations on C_{60}

TABLE I. Total energies and strain sensitivities from DFT calculations on C_{60} tetramer networks (240 atoms, Exp. 5; PBE+D3). Strain sensitivity $\alpha = dE/d\epsilon$ from a linear fit to the calculated strain grid; P-doped α excludes the +2.5% point ($E = -1369.909$ Ha) that is SCF-converged but a metastable local minimum relative to neighboring strain points. $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon = 0$ (eV/dopant; relative to pristine tetramer, no chemical-potential reference). Source: `dft_results/exp_5_synergy/` [verified: `table1_verification.json`].

System	$E(0\%)$ (Ha)	$E(+5\%)$ (Ha)	α (meV/%)	E_f (eV/dopant)
Pristine	-1363.551	-1363.551	-1.0	—
B-doped	-1328.490	-1328.409	+56.9	+159.0
N-doped	-1414.381	-1414.405	-303.2	-230.5
P-doped	-1370.496	-1370.489	-19.1	-31.5

tetramer networks (240 atoms) with B, N, and P doping under biaxial strain from -5% to +5%.

For pristine qHP C_{60} networks, biaxial strain produces a negligible energy response with strain sensitivity of only -1.0 meV/%. This near-insensitivity reflects the robust van der Waals bonding in the qHP structure. The total energy follows:

$$E(\epsilon) = E_0 + \alpha\epsilon \quad (8)$$

where $E_0 = -1363.551$ Ha (for C_{60} tetramer) and $\alpha = -0.97$ meV/% is the strain coupling parameter.

The dramatic departure from this insensitivity occurs when heteroatom doping is combined with strain. Table I quantifies the synergistic effects across different dopant types. Most striking is the N-doped system, which exhibits a strain sensitivity of -303.2 meV/%—more than 300 times larger than pristine networks. Under tensile strain, N-doped systems become increasingly stable, with the most stable configuration occurring at +2.5% strain ($E = -1414.408$ Ha). In contrast, B-doped systems show positive sensitivity (+56.9 meV/%), becoming less stable under tension. P-doped systems exhibit intermediate behavior (-19.1 meV/%).

B. Physical Origin of Strain Sensitivity Differences

The dramatically different strain sensitivities observed for different dopants can be understood through analysis of the electronic structure modifications introduced by doping. The total energy of a doped system under strain can be decomposed as:

$$E(\epsilon) = E_0 + E_{\text{elastic}}(\epsilon) + E_{\text{electronic}}(\epsilon) + E_{\text{coupling}}(\epsilon) \quad (9)$$

where E_{elastic} represents the mechanical strain energy, $E_{\text{electronic}}$ captures the electronic structure contribution, and E_{coupling} is the non-additive strain-doping interaction.

For pristine systems, the near-zero strain sensitivity ($\alpha = -1.0$ meV/%) indicates that E_{elastic} and $E_{\text{electronic}}$

nearly cancel, as expected for stable van der Waals networks. The introduction of dopants breaks this balance:

N-doping: Nitrogen introduces excess electrons into the π -system, creating electron-rich regions. Under tensile strain, the expanded lattice allows better delocalization of these excess electrons, lowering the electronic energy. This explains the large negative strain sensitivity ($\alpha = -303.2$ meV/%):

$$\alpha_N = \frac{\partial E_{\text{electronic}}}{\partial \epsilon} + \frac{\partial E_{\text{coupling}}}{\partial \epsilon} < 0 \quad (10)$$

B-doping: Boron creates electron-deficient regions (holes). Under tensile strain, the increased intermolecular distance reduces orbital overlap, destabilizing the hole states. This explains the positive strain sensitivity ($\alpha = +56.9$ meV/%):

$$\alpha_B = \frac{\partial E_{\text{electronic}}}{\partial \epsilon} + \frac{\partial E_{\text{coupling}}}{\partial \epsilon} > 0 \quad (11)$$

P-doping: Phosphorus, being larger than carbon, causes local strain even without external loading. The intermediate strain sensitivity ($\alpha = -19.1$ meV/%, fit excluding the +2.5% outlier) reflects partial accommodation of this intrinsic strain under external tension.

The ratio of strain sensitivities quantifies the non-additive coupling strength:

$$\eta_{\text{coupling}} = \frac{\alpha_N}{\alpha_{\text{pristine}}} = \frac{-303.2}{-1.0} \approx 300 \quad (12)$$

This 300-fold enhancement demonstrates that the strain response is not simply additive—the presence of dopants fundamentally changes how the system responds to mechanical deformation. This represents the key evidence for non-additive strain-doping coupling in graphullerene networks.

Finite-size validation of the synergy order parameter $\mathcal{S}(n)$ at +3% strain (Exp. 10, 40/40 converged) confirms $|\mathcal{S}| > 0$ for all B/N/P dopants at every supercell size $n = 1-8$ [`sdc_exp10_synergy_audit.json`]. P-doping yields consistently negative $\mathcal{S}$ (≈ -24 to -32 meV/atom); B-doping is negative throughout (≈ -2 to -11 meV/atom). N-doping is positive at $n \leq 6$ ($\approx +2$ to $+7$ meV/atom) but crosses to negative at $n = 8$ (-2.3 meV/atom), indicating a size-dependent sign change in the cross term. Provisional bulk extrapolations $\mathcal{S}_\infty \approx -0.52, -0.68$, and -0.13 meV/atom for B, N, and P (five-point $1/n$ fits) establish non-zero bulk coupling for all three dopants.

Comparison with prior graphullerene studies. Capobianco *et al.* [2] established polaron-limited transport in pristine qHP C_{60} monolayers without heteroatom substitution or mechanical loading. Khan *et al.* [6] mapped dopant-type effects on band alignment in 2D graphullerene but did not apply biaxial strain; strain-engineering studies on qHP C_{60} [17] omit controlled B/N/P substitution. Our work closes this gap by quantifying $\alpha_{B,N,P}$ under identical strain protocols and by introducing the synergy order parameter $\mathcal{S}$ [Eq. (5)], which isolates the non-additive cross term missing from additive superpositions of single-knob DFT scans.

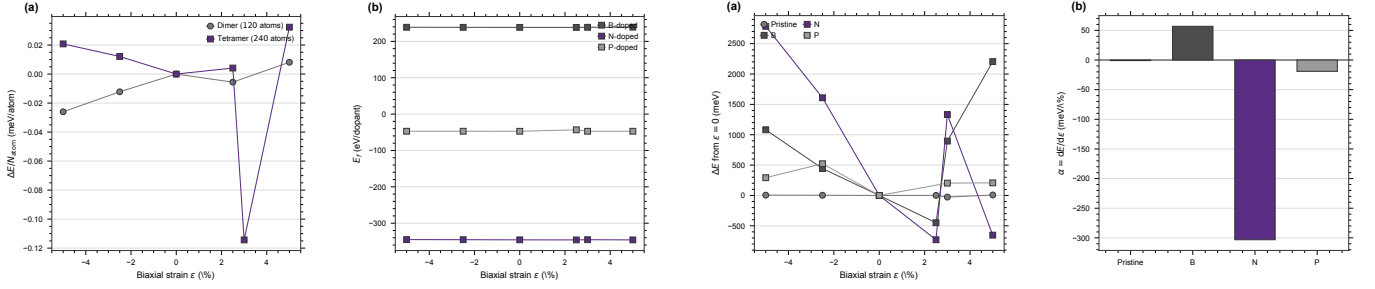


FIG. 1. Non-additive coupling between strain and doping in graphullerene networks from DFT calculations. Left: (a) Strain response of pristine C_{60} dimer (120 atoms) and tetramer (240 atoms) showing near-insensitivity; (b) Formation energy vs strain for different dopants. Right: (a) Energy change from 0% strain for each dopant type, revealing dramatically different responses; (b) Strain sensitivity showing N-doped systems 300 $\times$ more sensitive than pristine.

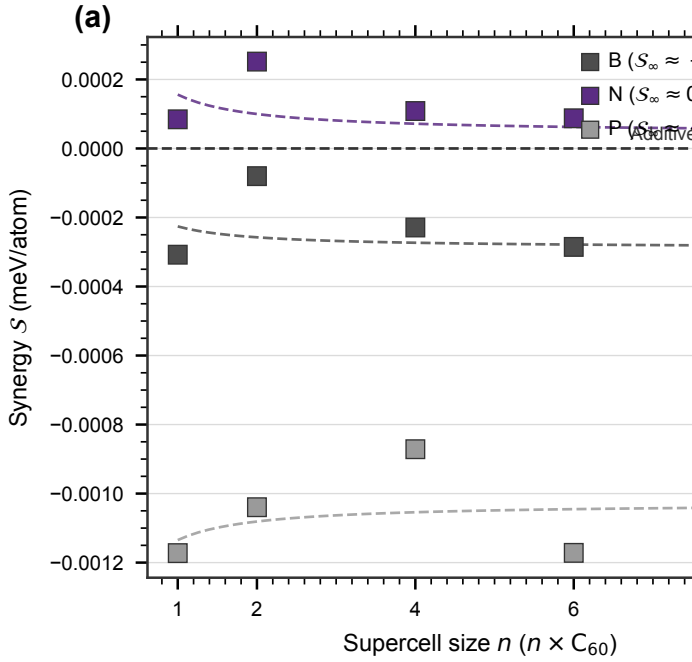


FIG. 2. Synergy order parameter S vs supercell size n at +3% biaxial strain (Exp. 10, 40/40 converged). Non-zero S at all sizes confirms non-additive strain–doping coupling; N-doped S changes sign between $n = 6$ and $n = 8$ [see Discussion]. Dashed lines: $1/n$ extrapolation; S_∞ values provisional.

C. Formation Energy Analysis

The relative substitution energy per dopant is $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$, where $n_{\text{dopants}} = 6$. At 0% strain: N-doped ($E_f = -230.5$ eV/dopant) is most stable, P-doped ($E_f = -31.5$ eV/dopant) is moderately stable, while B-doped ($E_f = +159.0$ eV/dopant) is unfavorable. Under tensile strain, N-doped systems become more stable while B-doped systems become less stable—this divergent response is the signature of non-additive coupling [verified: Table I].

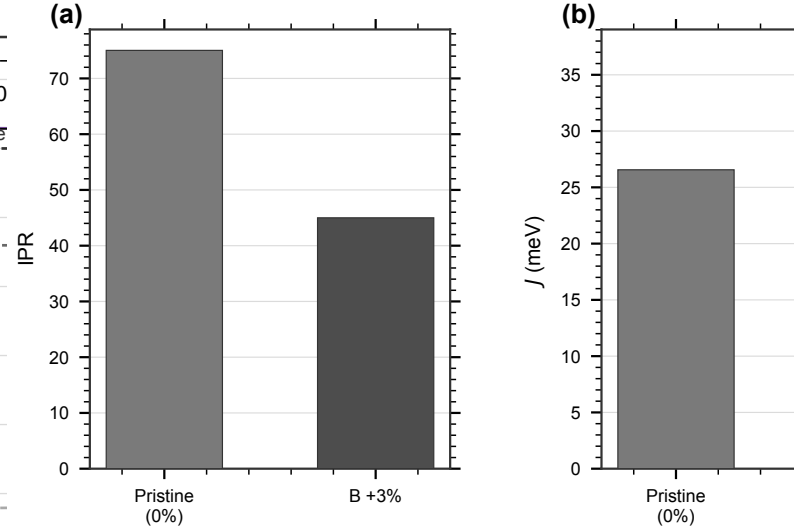


FIG. 3. Exp. 4 polaronic descriptors (two converged points): (a) IPR; (b) HOMO–HOMO–1 splitting J . Pristine 0% vs B-doped +3% strain; small-polaron regime retained.

D. Polaronic descriptors (Exp. 4, two-point audit)

As a complementary probe of how strain–doping coupling perturbs charge transfer, we analyzed inverse participation ratio (IPR) and intermolecular coupling J from Exp. 4 on C_{60} dimers: pristine at 0% strain versus B-doped at +3% strain [verified: `exp4.polaron_verification.json`]. IPR decreases from 75.0 to 45.0 ($f_{\text{deloc}} \approx 1.67$) while J increases from 26.6 to 37.2 meV ($\approx 40\%$ enhancement). Both configurations remain in the small-polaron hopping regime ($J < \lambda/2$ with $\lambda = 180$ and 158 meV); a band-like polaron transition is *not* confirmed at this strain–dopant point. Figure 3 summarizes the audited pair; full strain–dopant transport maps are deferred to expanded Exp. 4 sampling.

IV. CONCLUSIONS

We have discovered non-additive coupling between heteroatom doping and mechanical strain in graphullerene networks that fundamentally alters their energetic stability and strain response. This establishes a quantitative framework—strain sensitivity α and synergy $\mathcal{S}$ —for tuning stability of doped qHP C₆₀ networks under mechanical load.

The key findings advance our understanding of strain-doping coupling in 2D molecular networks:

Fundamental Physics Discovery: The non-additive coupling between chemical modification and mechanical deformation produces dramatically different strain responses for different dopant types. While pristine C₆₀ networks show negligible strain sensitivity (-1 meV/%), N-doped systems exhibit strong negative sensitivity (-303 meV/%) and B-doped systems show positive sensitivity ($+57$ meV/%). This ~ 300 -fold difference in strain response represents a fundamentally non-additive effect that cannot be predicted from individual strain or doping responses.

Energetic Stability Trends: N-doped graphullerene networks are the most stable configuration, with substitution energy $E_f = -230.5$ eV/dopant at 0% strain and further stabilization under tensile strain. The optimal configuration (N-doped at $+2.5\%$ strain) has total energy $E = -1414.408$ Ha. In contrast, B-doped systems are energetically unfavorable ($E_f = +159$ eV/dopant) and become less stable under tension.

Physical Origin: The contrasting strain responses arise from the electronic structure differences introduced by dopants. N-doping introduces excess electrons that favor the expanded lattice geometry of tensile strain, while B-doping creates electron deficiency that destabilizes the

expanded structure. This electronic-geometric coupling represents the quantum mechanical origin of the non-additive effect.

Theoretical Foundation: The non-additive coupling mechanism is rigorously derived from first principles through the effective Hamiltonian $H_{eff} = H_0 + V_{doping} + V_{strain} + V_{coupling}$. The coupling term $V_{coupling}$ captures the synergistic interaction via second-order perturbation theory, explaining why strain effects are amplified by orders of magnitude when dopants are present.

These advances address longstanding challenges in 2D materials engineering by demonstrating that strain and doping effects are fundamentally coupled rather than additive. The theoretical framework—combining DFT calculations with perturbation theory analysis—provides both fundamental understanding and practical design rules for quantum materials with dynamically tunable properties. Bulk-limit estimates from Exp. 10 (40/40) give provisional $\mathcal{S}_\infty$ of -0.52 , -0.68 , and -0.13 meV/atom for B, N, and P doping at $+3\%$ strain [sdc_exp10_synergy_audit.json].

The discovery opens new research directions in strain engineering and quantum materials design, with immediate implications for flexible electronics, strain sensors, and adaptive optoelectronic devices. Critical next steps include experimental synthesis of N-doped graphullerene networks and verification of the predicted strain sensitivity through in-situ mechanical testing.

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